

Mapping the First Hop: Denominators, Hierarchy, and Gateway Structure in Reported Indian Foreign Ownership Networks

Canonical research synthesis

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Abstract

A registered foreign investment is the first hop of a corporate structure, not the structure itself. This paper reconstructs what sits beneath the first hop for 28 Indian parent buckets in a supplied research dataset. The source is not a firm cross-section: 3,742 candidate source rows reduce to 3,567 preferred target-years, 1,834 structural target occurrences, 1,830 parent-scoped normalized entity candidates, and 1,818 global normalized candidates. Keeping these denominators separate changes both rankings and interpretation. The 186 mapping channels in the file expand into 1,834 foreign targets—a mean of 9.9 and a median of 2 targets per channel—and 47.5% of targets sit two or more reported levels below the Indian parent. That structure is highly concentrated. The modal channel contains 59.7% of targets pooled and 62.1% under equal-parent weighting; among 24 sufficiently large groups the median largest observed subtree contains 43.2% of normalized entities, and 41.7% under a strict directed-acyclic-graph dominator definition. How much structure follows also depends on where the first hop lands: Dutch first-hop entities have a mean of 34.8 downstream entities (parent-equal mean 37.1, leave-one-parent-out 21.2–39.7) against 9.9 for the much more numerous United States first hops. Direct relations are also commonly international: 951/1,650 nonroot edges (57.6%) cross a jurisdiction boundary, with an equal-parent estimate of 58.7%. These results survive entity deduplication, parent balancing, leave-one-parent-out checks, and alternative network definitions. Other apparent patterns do not: reported level and reconstructed graph distance disagree for 236/1,687 complete paths, and pooled depth, sector, ownership, and financial patterns are driven by parent composition or measurement limits. The paper therefore establishes a measurement-corrected topology for this dataset, and separates it explicitly from the legal, tax, and regulatory interpretations that would require external evidence.

Keywords: ownership networks; corporate hierarchy; subsidiary geography; entity resolution; robustness; denominator audit

Scope note. Every quantitative claim below is produced by the pipeline in this repository from one immutable input, and every number in the text is a generated macro rather than a transcribed figure. The paper carries no literature review and no institutional or regulatory interpretation: those would require independently sourced evidence, which is deliberately outside the verified base used here. Table 7 states that separation, and Section 10 lists what a sourced revision would need to settle.

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1 Introduction

When a firm registers a foreign investment, what gets recorded is one company in one country. What the firm actually owns is usually a structure: the registered company holds other companies, which hold others in turn, often in different jurisdictions. Any measure built on the registered entity—an aggregate of outward investment, a valuation of a foreign affiliate, an index of corporate complexity—therefore measures the top of something whose size and shape are not observed at the point of registration.

This paper asks how large and how concentrated that hidden structure is, using a supplied research dataset that reconstructs reported ownership below the first foreign hop for 28 Indian parent groups. Three facts frame the exercise. The 186 registration or mapping channels in the file expand into 1,834 reported foreign targets, a mean of 9.9 and a median of 2 targets per channel. 47.5% of those targets sit two or more reported levels below the Indian parent, so the majority of what the first hop stands for is not the first hop. And the structure below is not a broad tree but a narrow one: for the median large group in this sample, a single observed company sits above roughly two-fifths of the group’s entire foreign network.

Answering that question well is mostly a measurement problem, and the paper treats it as one. Datasets of this kind combine objects that look interchangeable but are not: source records, reporting years, registered investments, named subsidiaries, parent relations, and complete ownership paths. Treating any one of them as “the number of firms” mechanically changes which parent groups and which jurisdictions dominate a result. Hierarchy adds a second trap. A location can be prominent because many entities reside there, because a few entities there have many descendants, or because one unusually large group repeats one route many times. Those are three different empirical statements, and only the second is about intermediation.

The central question is therefore bounded deliberately: *how do the parent groups in the supplied 28-parent dataset organize their reported foreign ownership networks across mapping channels, hierarchy, and jurisdictions, and which apparent regularities survive correcting units, duplicates, parent weights, missing nodes, and graph definitions?* The goal is measurement-corrected description of these groups. The data do not identify why a structure exists, whether it is current, what legal function a named entity performs, or whether any route serves a tax or regulatory purpose.

The analysis makes five contributions within that boundary. First, it builds an explicit denominator ladder from source observations to target, entity, edge, path, mapping-channel, and parent units, and shows that the choice reverses parent rankings. Second, it reconstructs a parent-scoped directed ownership graph that retains named-but-unscraped parents together with their supplied countries, rather than discarding the only information those nodes carry. Third, it separates three jurisdiction roles that are routinely conflated: residence, unique intermediary presence, and descendant-weighted path exposure. Fourth, it reports pooled, equal-parent, leave-one-parent-out, and alternative-definition estimates side by side for every headline number, so that a reader can see which results are properties of the network and which are properties of the largest parent group. Fifth, it treats failed hypotheses as results: several initially striking depth, sector, ownership, and financial relationships do not survive the relevant correction, and the paper reports them as negative findings rather than dropping them.

Four substantive results follow. Parent architectures are heterogeneous along axes that cannot be collapsed into one defensible complexity ranking: the largest network in the sample is reported to reach level 12, another network less than half its size reaches the same level, and a third with almost two hundred entities never goes below level 2. Networks are concentrated behind a dominant mapping channel or one large observed subtree, and that concentration is similar under four different definitions of “dominant.” Where the first hop lands predicts how much structure sits below it:

Dutch first hops have a mean of 34.8 downstream entities against 9.9 for the twice-as-numerous United States first hops, and the ordering survives parent balancing and omission of any single parent. Finally, a majority of reported child–parent relations below the first hop cross a jurisdiction boundary (57.6%), which is the single most definition-stable statistic in the study.

Three results in the opposite direction matter as much. Reported hierarchy level and reconstructed graph distance disagree for 236/1,687 complete paths, and one parent group supplies 88.1% of the disagreements, so “depth” is not one variable. An apparent sector–depth pattern is two groups. An apparent tendency for ownership stakes to be more complete deeper in the hierarchy is parent composition. And the financial layer is sparse, unit-less, and partly built from reused extracted evidence, so it supports an audit of data quality rather than any comparison of firm performance.

Throughout, the paper distinguishes what this dataset establishes from what it cannot. Unusual topology is not evidence of wrongdoing, a country label is not a statement about tax residence or entity function, and a maintained list of “financial centres” is an imposed classification rather than an observed variable. Table 7 states that separation explicitly, and the paper is written to respect it.

The remainder proceeds as follows. Section 2 defines the units and Section 3 the graph, paths, and estimands. Section 4 documents the denominator correction and architectural heterogeneity; Section 5 the expansion of mapping channels and the concentration of networks behind dominant channels and subtrees; Section 6 the geography of intermediation. Section 7 confronts the two depth measures, Section 8 audits the ownership and financial fields, and Section 9 collects the robustness evidence. Section 10 states the interpretation boundary and the questions this file cannot settle.

2 Data and analytical units

2.1 Source structure and scope

The immutable input is a Stata file containing candidate financial-statement sources attached to reported foreign targets. The file has 3,742 rows and 122 standardized jurisdiction labels across 28 supplied ultimate-parent buckets. Rows are not unique companies. A target can recur for multiple fiscal years and can have multiple candidate sources within a year. The upstream preferred-source flag selects 3,567 target-year rows, but structural fields are invariant within each target identifier. The structural cross-section therefore contains 1,834 target occurrences.

The sample is not accompanied by a sampling frame for all Indian outward investment. No result below is generalized beyond the supplied parent buckets. Similarly, the file provides reported names and ownership relationships, not a complete legal-entity registry or verified historical ownership charts.

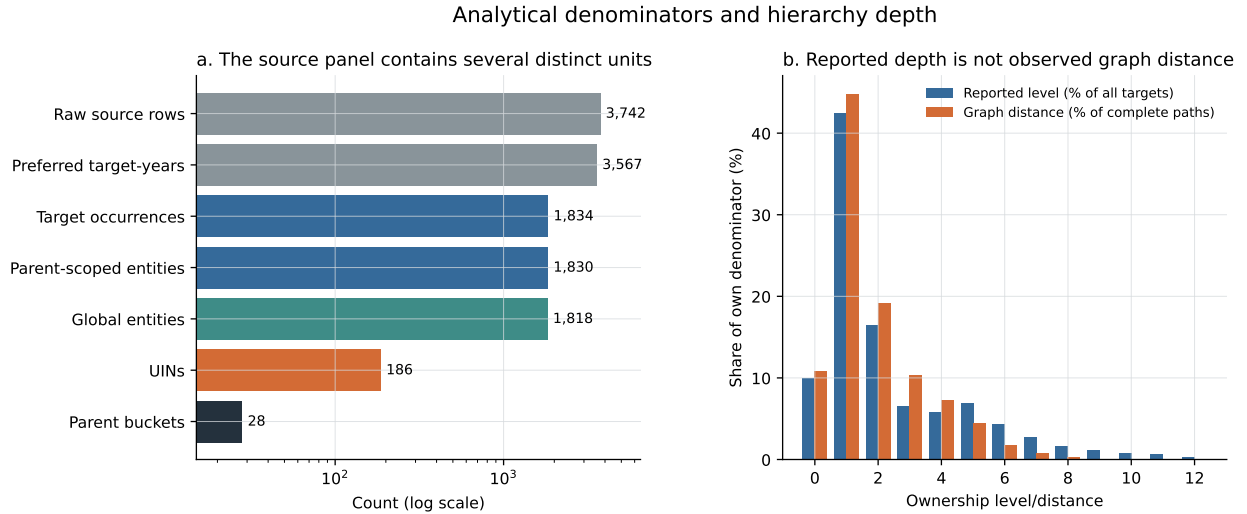
2.2 Targets, normalized entities, and mapping channels

The primary path and exposure unit is the target identifier. For within-parent architecture, entity names are normalized conservatively: Unicode is reduced to ASCII, case and punctuation are removed, ampersands are standardized, and the supplied country is spelling-harmonized. A parent-scoped entity key is the combination of ultimate-parent bucket, normalized name, and standardized country. This yields 1,830 candidates, only slightly fewer than the 1,834 targets. Dropping the parent component yields 1,818 global candidates. The difference is small in aggregate, but the estimands are conceptually distinct: a shared venture may be one global entity while still representing exposure in more than one parent network.

Table 1: The canonical denominator ladder

Analytical unit	Count	Definition
Raw source rows	3,742	source candidate
Preferred target-year rows	3,567	$\text{target_id} \times \text{fiscal_year}$
Structural target occurrences	1,834	target_id
Parent-scoped normalized entity candidates	1,830	$\text{parent} \times \text{normalized name-country}$
Global normalized entity candidates	1,818	$\text{normalized name-country}$
UINs	186	uin
Observed level-0 targets	184	target occurrence
Ultimate-parent buckets	28	source parent label
Complete observed paths	1,687	target path

Notes: Counts are generated from the immutable source by the canonical pipeline. A target occurrence is one reported ownership-path target. Normalized entities are conservative name-country candidates, not verified legal identities. A complete observed path reaches the Indian parent through observed entity nodes.

**Figure 1:** Six analytical units, and two depth measures on their own denominators

Notes: Panel (a) uses a logarithmic count scale because the units differ sharply in size. Panel (b) compares the supplied reported level for all targets with graph distance among complete observed paths; these are not interchangeable measures.

The file also contains 186 UIN values. They are retained as supplied and treated only as registration or mapping channels. The analysis does not infer the semantics of their substrings, a registration date, or legal-entity identity. 184 UINs have an observed level-0 target, so two mapping channels lack an observed first-hop entity in this file.

2.3 Financial observations

Financial variables remain at the preferred target-year level. Coverage gates are reported separately from network construction. A target has a located source if any source row carries the supplied source-found indicator, a parsed balance sheet if equity is numeric in any row, and a valuation-ready record if the upstream ready flag is present at least once. Absolute monetary values are not pooled across currencies or unspecified units.

3 Ownership-network construction and estimands

3.1 What the hierarchy is

It helps to fix the object before measuring it. Each parent group is rooted at one Indian company. Directly beneath it sit the first foreign hops, recorded at level 0: these are the companies the Indian parent is reported to hold directly. A level-1 target names a level-0 company as its immediate parent, a level-2 target names a level-1 company, and so on. A group’s network is therefore a set of chains

$$\text{Indian parent} \rightarrow \underbrace{E_0}_{\text{level 0}} \rightarrow E_1 \rightarrow \cdots \rightarrow E_k,$$

where each arrow is a reported immediate-parent relation and each E_j carries its own country label. Nothing constrains consecutive entities to share a country: an Indian parent may hold a Dutch company that holds a German company that holds a Brazilian one. Nothing constrains the shape either. A group can be wide and flat, with many level-0 entities and no further layers, or narrow and deep, with one level-0 entity above a chain that continues for ten more steps. Which of those a group looks like is the architectural variation this paper measures.

Two properties of that structure drive the rest of the analysis. First, descendants accumulate: an entity high in a chain has every entity below it in its subtree, so counting “entities below jurisdiction X ” counts a shared ancestor once for each descendant. Second, the same target can be counted as an occurrence, as a normalized entity, or as a path, and those three counts differ. The estimands defined below keep them apart.

3.2 Edge resolution

Each structural target supplies an entity name and country, reported level, immediate-parent name and country, UIN, and ultimate-parent bucket. Candidate parents are formed within the same ultimate-parent bucket from conservative name aliases. Candidate matches are ranked lexicographically by whether they occupy the expected preceding level, whether the supplied parent country matches, whether the original names match exactly, and whether the UIN matches. The canonical build accepts no fuzzy match.

The result is one occurrence-level edge for each target. Level-0 targets link to an explicit Indian ultimate-parent node. Below level 0, 1,500 of 1,650 relations link to an observed entity at the expected prior level, 72 link to an observed entity at another reported level, and 78 terminate at one of 18 normalized named-but-unscrapped parent nodes. These unresolved nodes retain their

supplied country. Discarding the country would understate upstream exposure for some groups; treating the node as observed would overstate path completeness.

Occurrence edges are collapsed to a logical entity graph for subtree and gateway calculations. The parent-specific normalized graphs are directed and acyclic under the accepted links. The graph preserves alternative reported parents where they exist rather than forcing a tree.

3.3 Paths and depth

For every target, recursion follows observed parents toward the ultimate parent. A path is complete only when it reaches that root through observed entities. This succeeds for 1,687/1,834 targets (92.0%). A path that encounters a named missing node remains useful for country exposure up to that node but is not declared complete.

The supplied reported level and reconstructed graph distance are stored separately. Reported level may encode a hierarchy with nodes missing from the target table. Conversely, cross-channel name matching can create a shorter observed route than the reported sequence. Without external ownership charts, neither measure is relabeled as “true” depth.

3.4 Network concentration

Four concentration quantities answer different questions.

- The *modal-UIN share* is the fraction of a parent’s targets assigned to its largest UIN channel.
- The *largest observed subtree* counts normalized entity descendants below the observed entity with the largest descendant set, divided by the other entities in that parent network.
- The *named-node-inclusive subtree* also allows an unresolved named parent to be the top node, while still not calling the path complete.
- The *strict DAG dominator* removes an observed node and counts the entity nodes no longer reachable from the ultimate parent. It therefore respects alternative routes that bypass the candidate node.

The subtree summaries use the 24 parent groups with at least fifteen targets. That threshold prevents very small groups from determining a median while retaining most of the structural sample. Subtree language describes the mapped topology. It does not establish legal control, cash-flow dependence, or an operational chokepoint.

3.5 Geographic roles

The analysis distinguishes three jurisdiction roles. *Residence* counts normalized entities located in a jurisdiction. An *observed intermediary* is a unique observed entity that appears as the parent of at least one logical edge. A jurisdiction is *strictly upstream* of a target if it appears above the terminal target in the reconstructed path; the target’s own residence is excluded from its exposure set. India is excluded as the ultimate root in foreign-jurisdiction role and cover calculations.

A *gateway* is an observed level-0 entity. Gateway amplification is the number of normalized entity descendants below it. Jurisdiction means weight gateways; parent-equal means first average gateways within parent and then average parents. A robustness subset retains gateways for which no single direct-child branch contains a majority of descendants.

An edge is cross-border when the standardized child and immediate-parent country labels differ. The cross-border statistic excludes level-0 links to India, which would otherwise be cross-border by construction.

3.6 Weighting and robustness

For a binary target outcome y_i and parent $p(i)$, the pooled estimand is $N^{-1} \sum_i y_i$. The equal-parent estimand is

$$\frac{1}{P} \sum_{p=1}^P \left(\frac{1}{N_p} \sum_{i:p(i)=p} y_i \right).$$

The former describes target-weighted exposure in this file; the latter gives each parent bucket the same weight. Leave-one-parent-out estimates repeat the relevant pooled or fixed-effect calculation after excluding each parent in turn. Where units differ—entities, edges, paths, gateways, or parents—the paper labels the denominator explicitly rather than comparing percentages as though they shared an estimand.

4 Denominators and parent architecture

4.1 Raw-row weighting changes the sample composition

The denominator correction is empirically first order, and the cleanest demonstration is that it reverses a ranking. Tata Communications contributes 550 source rows but only 49 parent-scoped normalized entities. Motherson contributes 332 rows but 309 normalized entities. The former is therefore the largest group in the file by source rows and roughly a sixth the size of the latter by network entities. Nothing about the corporate structures changed between those two sentences; only the unit did. Any statistic computed directly on source rows would overweight targets with longer or more duplicated financial-source histories.

The closeness of the target and normalized-entity totals is reassuring but does not make the units interchangeable. Four parent-scoped alias collapses are enough to matter for exact graph topology, and cross-parent shared names have a different interpretation from within-parent duplicates. The canonical tables therefore retain target and entity counts side by side.

4.2 Architecture is multidimensional

Figure 2 shows that group size, geographic breadth, reported depth, and channel concentration do not move together. Table 2 gives illustrative contrasts. Motherson is the largest entity network in the sample and has the greatest reported maximum depth, while Wipro spans more jurisdiction labels. Reliance Industries is large but shallow in reported levels. Tata Communications is much smaller as an entity network than its source-row count suggests. ONGC Videsh has many UIN channels relative to its target count and a shallow observed graph.

These comparisons do not support a one-dimensional complexity index. A group can be broad but shallow, deep but concentrated behind one entry channel, or large because one subtree contains many acquired or portfolio entities. Calling any one of those structures “more complex” would impose an unvalidated weighting across substantively different axes.

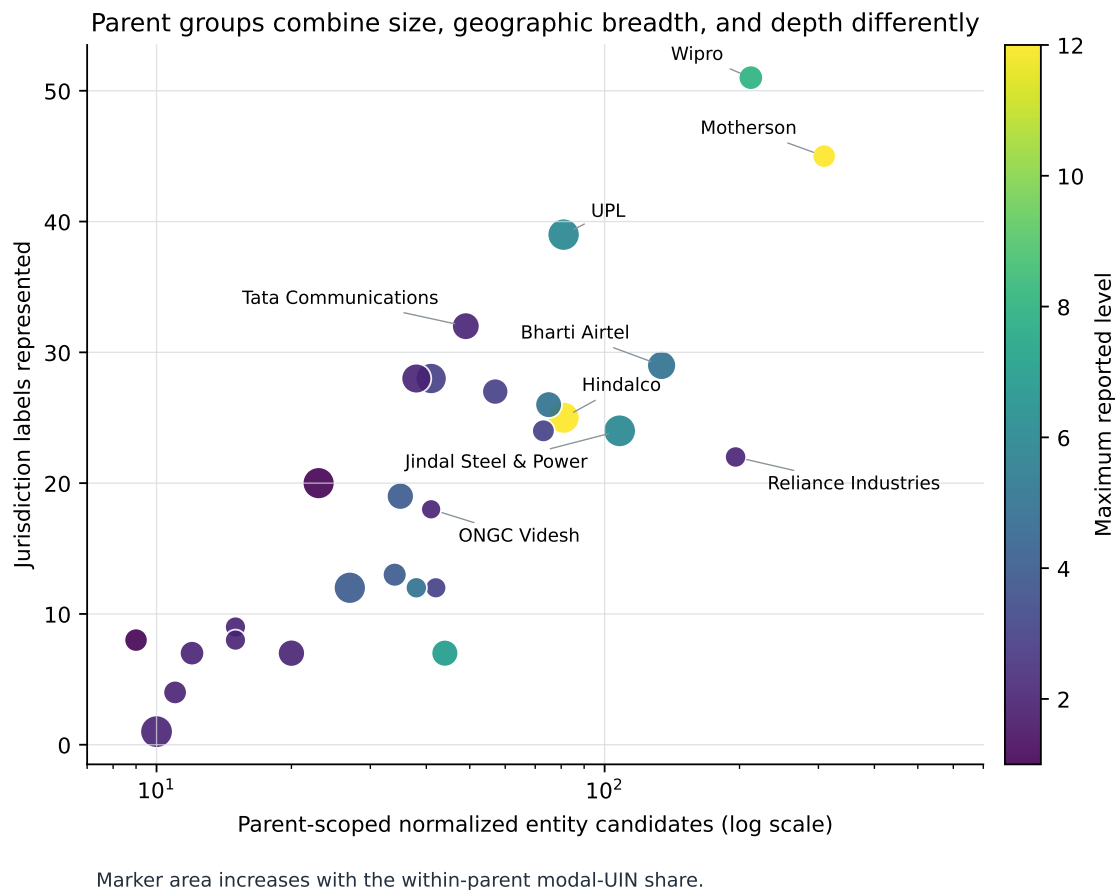


Figure 2: Size, jurisdictional breadth, reported depth, and channel concentration by parent group
Notes: Each marker is one supplied ultimate-parent bucket. Marker color is maximum reported level; marker area increases with the within-parent modal-UIN target share. Entity count uses the parent-scope normalized name-country key.

Table 2: Illustrative parent architectures

Parent bucket	Raw rows	Entities	Countries	Reported max	Graph max	Modal UIN (%)	Complete paths (%)
Motherson	332	309	45	12	8	44.7	100
Wipro	504	212	51	8	7	50.2	88.7
Reliance Industries	277	196	22	2	2	34.7	100
Hindalco	81	81	25	12	6	95.1	46.9
Tata Communications	550	49	32	2	2	68	100
ONGC Videsh	41	41	18	2	2	29.3	100

Notes: Graph maximum is reconstructed distance among complete paths. Modal UIN is the largest UIN's share of target occurrences. Complete paths are reported as a percentage of the parent's targets. Rows are selected to illustrate contrasting architectures, not to rank firms.

5 Dominant channels and large subtrees

5.1 How far one mapping channel reaches

Section 4 showed that parent groups differ. This section asks what a single registration or mapping channel stands for, and how much of a group’s network hangs from one place.

The 186 channels in the file resolve to 1,834 reported foreign targets: a mean of 9.9 targets per channel, but a median of only 2. The two statistics differ because the distribution is extremely skewed. 89 of 186 channels (47.8%) resolve to exactly one target, while the ten largest channels together account for 49.2% of all targets and the single largest contains 138. 184 of 186 channels have an observed level-0 target, so two channels lack an observed first foreign hop in this file, and none has more than one.

Two things follow, and they must be kept apart. As a statement about this mapping, the expansion factor is large and unevenly distributed: for half of the channels the first hop is the whole of what is recorded, while a handful of channels carry most of the entity mass. As a statement about registration practice or official statistics it establishes nothing, because the file contains no evidence on how these channels were assigned, what their identifiers encode, or what share of any register they represent. The paper uses the expansion only as a description of the supplied mapping.

5.2 Concentration behind one channel or one subtree

The largest UIN within each parent accounts for 1,095/1,834 targets, or 59.7% pooled. Giving every parent equal weight produces 62.1%; the median parent is 61.2%, and the pooled leave-one-parent-out interval is narrow (Figure 8). Channel concentration is therefore not only an artifact of the largest parent bucket. At the same time, UIN is an administrative or mapping unit in this analysis, not a verified company or control node.

Graph concentration yields a complementary result. Across the 24 larger groups, the median largest observed subtree contains 43.2% of the other normalized entities. The median changes only modestly to 41.7% under the strict DAG dominator definition, which allows alternative paths to bypass the removed node. When named-but-unscraped parent nodes may sit at the top of a subtree, the median rises to 50.8%.

The shape of that concentration is layering rather than branching. Of the 1,830 normalized entities, 78.0% hold nothing at all, and of the 403 that hold something, 198 (49.1%) hold exactly one entity. Wide branching is confined to a few nodes: the 34 entities with ten or more direct children originate 742 of the 1,572 logical edges that begin at an observed entity (47.2%). A network can therefore be deep without being bushy, which is why maximum depth and entity count are separate axes in Section 4. Two of the three largest hubs carry names that identify them as investment funds, so a high out-degree is not by itself evidence of an operating span of control; the robustness matrix reports the three-country exposure with those funds’ children removed.

Figure 3 makes the conceptual separation visible. UIN and subtree shares are often close, but not always. A mapping channel can contain many targets whose observed graph is split, while a linked subtree can span more than one UIN. Missing parents can also reverse the apparent concentration of a particular group. The defensible result is that dominant mapped channels and large observed subtrees are common in this sample. It is not that every network has one legally dispositive company.

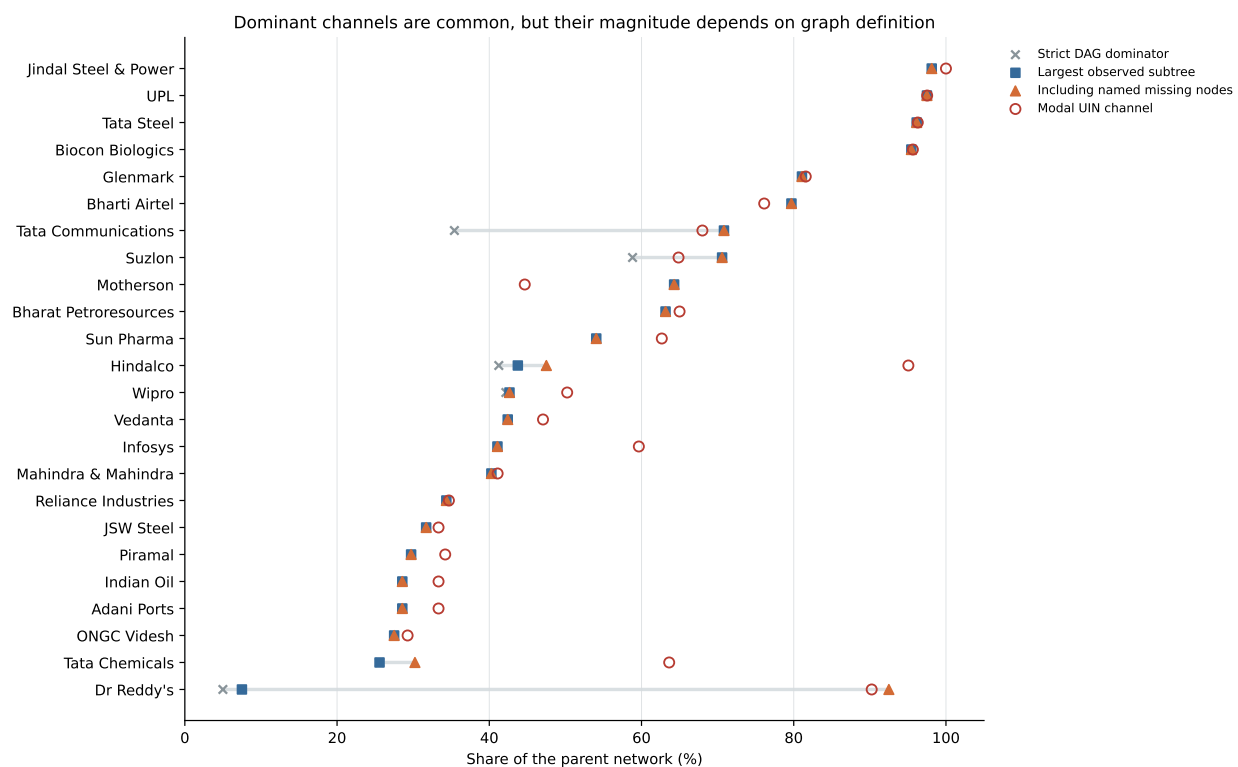


Figure 3: Channel concentration under alternative graph definitions

Notes: The sample is parent groups with at least fifteen targets. Subtree shares use normalized entity candidates and exclude the top node from the denominator. The strict DAG measure counts entities detached from the ultimate root after removing the candidate node. The hollow marker is the target-weighted modal-UIN share and therefore uses a different unit.

6 Geography and intermediary structure

Concentration says how much of a network hangs from one node. This section asks where those nodes are. The order of the evidence matters: the paper first establishes that layering is international at all, then that first-hop location predicts how much structure follows, and only then compares jurisdictions — by which point it is clear that “common” and “upstream” are different properties.

6.1 Cross-border ownership relations

Among 1,650 reported child–parent relations below level 0, 951 cross a standardized jurisdiction boundary. The pooled share is 57.6%, the equal-parent mean is 58.7%, and the leave-one-parent-out range is 56.1–61.8%. Recomputing with the raw supplied country labels gives essentially the same result (Table 6). This is one of the most definition-stable structural facts in the study.

The interpretation remains modest. A cross-border edge says only that the reported child and immediate parent carry different country labels. It does not reveal operating location, management, tax residence, legal control, or the purpose of the relation.

6.2 First-hop gateway amplification

Table 3 and Figure 4 treat each observed level-0 entity as a gateway and count normalized descendants below it. Dutch gateways have a mean of 34.8 descendants and median of 12. The parent-equal mean is 37.1, and the pooled leave-one-parent-out means range from 21.2 to 39.7. Restricting to gateways without a single majority child branch gives a mean of 37.8, so the average is not generated only by one immediate acquisition branch per gateway.

Mauritian gateways also show greater amplification than the United States comparison (mean 17.9, median 5), although their median is well below their mean. The contrast with the United States is the sharpest in the table: 28 United States gateways, more than twice the Dutch count and spanning as many parent groups, carry a mean of only 9.9 descendants and a median of 1.5. Being a frequent first destination and being a position from which a network extends are, in this sample, different things.

Three properties keep this from being one group’s architecture. The 13 Dutch gateways span 12 different parent groups; the parent-equal mean exceeds the gateway-weighted mean, so balancing groups strengthens rather than weakens the pattern; and the ranking survives dropping any single parent. The magnitudes, however, are estimand-specific. The Dutch-to-United-States ratio is roughly three and a half on means and about eight on medians, because the United States distribution is dominated by single-entity first hops. Any statement of the form “ X times more” must name which of the two it uses.

Interpretation stops there. A first hop with many descendants may be an acquisition whose target already owned a large group, a financing or governance vehicle assembled afterwards, an operating headquarters, or an artifact of how the mapping attributes intermediate layers. This file observes none of those histories, and the descendant count cannot distinguish them.

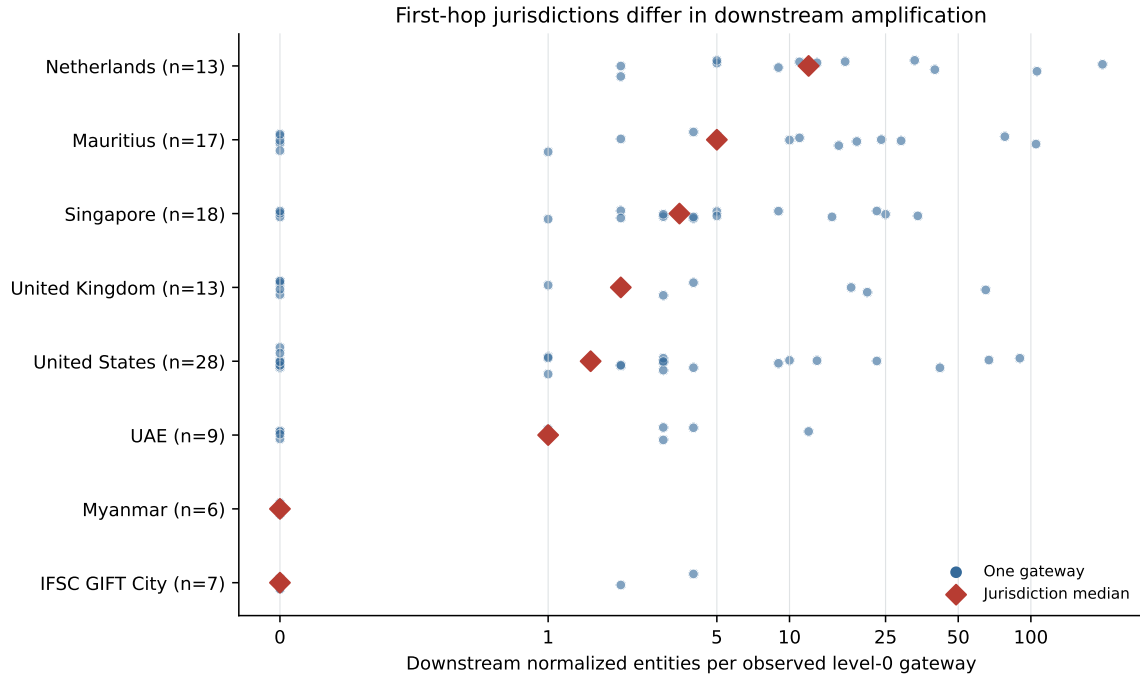
6.3 Residence, intermediary prevalence, and path exposure

There are 403 unique observed parent nodes in the logical entity graph. Table 4 contrasts their distribution with resident entity counts and target paths. The United States is the largest resident location among the displayed jurisdictions. The Netherlands and Mauritius, however, have substantially more target paths below them relative to their resident counts. Figure 5 generalizes that

Table 3: Downstream amplification by selected first-hop jurisdiction

Gateway jurisdiction	Gateways	Parents	Mean	Median	Parent-equal mean	LOO mean range
Netherlands	13	12	34.8	12	37.1	21.2 to 39.7
Mauritius	17	11	17.9	5	19.7	12.4 to 19.5
United States	28	12	9.9	1.5	9.2	7.2 to 11.3
Singapore	18	12	7.5	3.5	7.4	6.0 to 8.5
United Arab Emirates	9	6	2.6	1	2.1	1.6 to 2.9

Notes: One gateway is one observed level-0 normalized entity. Mean and median weight gateways. The parent-equal mean first averages gateways within parent. The LOO range repeats the gateway-weighted mean after omitting each parent. The table shows selected jurisdictions; the full generated table contains every jurisdiction with at least three gateways.

**Figure 4:** Downstream entities per observed first-hop entity, by jurisdiction

Notes: Each blue point is an observed level-0 gateway; diamonds mark jurisdiction medians. The horizontal axis uses a symmetric-log display to retain zero-descendant gateways. The plotted jurisdictions each have at least five observed gateways.

Table 4: Selected jurisdiction roles under distinct denominators

Jurisdiction	Residents	Intermediaries	Paths below	Pooled (%)	Parent-equal (%)	Parents
United States	363	69	381	20.8	14.9	16
Netherlands	110	59	540	29.4	20.9	16
Mauritius	57	25	308	16.8	14.7	10
United Kingdom	99	39	219	11.9	10.2	16
Singapore	60	21	147	8.0	13.6	13

Notes: Residents are parent-scoped normalized entities. Intermediaries are unique observed parent nodes. Paths below count target paths on which the jurisdiction appears strictly upstream, including a supplied country on a named unresolved node. Pooled and parent-equal percentages use target paths; “Parents” is the number of parent buckets affected.

comparison: points above the diagonal have more descendant paths than resident entities, while points below it are more terminal in this mapped topology.

The Netherlands is strictly upstream of 29.4% of target paths, compared with 20.9% after equal-parent weighting. The pooled value is therefore partly a function of the sizes of the parent networks using Dutch routes, but the role does not disappear when parents are balanced.

The greedy union of the Netherlands, United States, and Mauritius is strictly upstream of 1,140/1,834 target paths (62.2%). The equal-parent union is only 46.7%, and removing a large set of venture-fund children gives 59.8%. The gap is substantive: the pooled cover is a descendant-weighted exposure statistic, not the share of unique intermediaries and not a counterfactual measure of what would be disconnected if three jurisdictions vanished.

6.4 Why a fixed “centre” category is not a core result

A maintained list of frequently discussed financial or intermediary locations is upstream of 61.8% of target paths, and the equal-parent estimate is 57.2%. But only 38.0% of unique observed intermediary nodes fall in the same list. The path figure repeats a shared upstream node for every descendant, and the list itself is imposed rather than observed. It also combines locations that may host operating, financing, acquisition, or holding entities. The canonical study therefore reports specific measured roles and keeps the aggregate classification as a labeled sensitivity analysis.

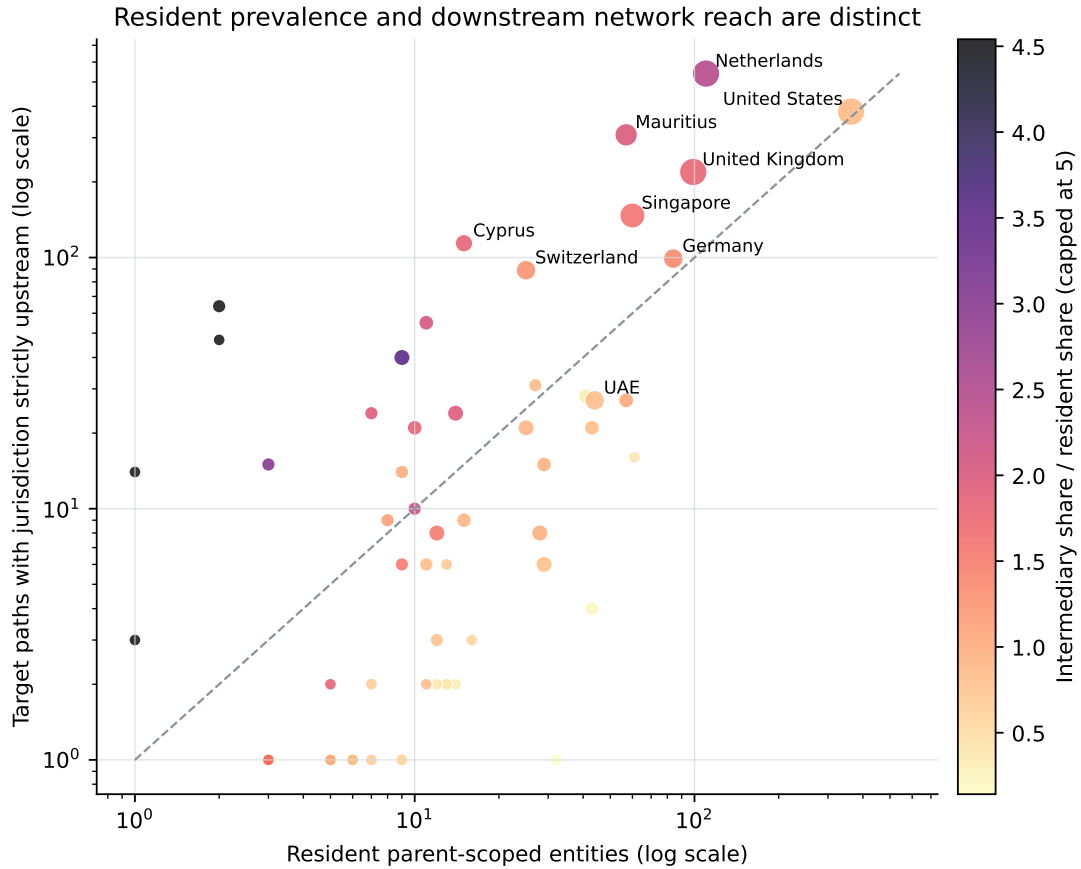


Figure 5: Resident entities against target paths with the jurisdiction strictly upstream
Notes: Both axes are logarithmic. Marker area increases with the number of parent groups affected. Color is the jurisdiction's share of unique observed intermediary nodes divided by its share of resident entities, capped for display. India is excluded as the ultimate root. Only jurisdictions appearing on the paths of at least five parent groups are labeled: several unlabeled points in the upper left have very high paths per resident entity but rest on one or two groups, and are not general patterns.

7 Reported depth and observed graph distance

The previous sections used depth loosely, as the number of steps between the Indian parent and a target. This section shows why that word needs a qualifier in this dataset, and why the deep tail in particular should not be read as a sample-wide feature.

Nearly half of targets are reported below the first foreign layer: 871/1,834 (47.5%) have reported level 2 or higher. The apparently deep tail is less stable. Reported level 5 or higher contains 342 targets (18.6%), while the equal-parent mean is only 7.3%. Among complete paths, just 7.4% have reconstructed distance 5 or greater.

The measures disagree directly for 236/1,687 complete paths (14.0%). Motherson contributes 88.1% of these mismatches. Figure 6 shows that the maximum reported and reconstructed depths can diverge sharply within a group. Reported level may preserve omitted layers; reconstructed distance may be a lower bound. The concentration of disagreements in one parent also means that a pooled depth correction cannot be assumed to apply uniformly.

The sector result provides a falsification exercise. Of 508 targets labeled general manufacturing, 242 are reported at level 5 or deeper. That looks like a broad sector-depth association until parent composition is examined: Motherson and Hindalco account for 240/242 (99.2%) of the deep manufacturing targets. The paper therefore rejects “manufacturing subsidiaries are generally deeper” as a sample-wide conclusion.

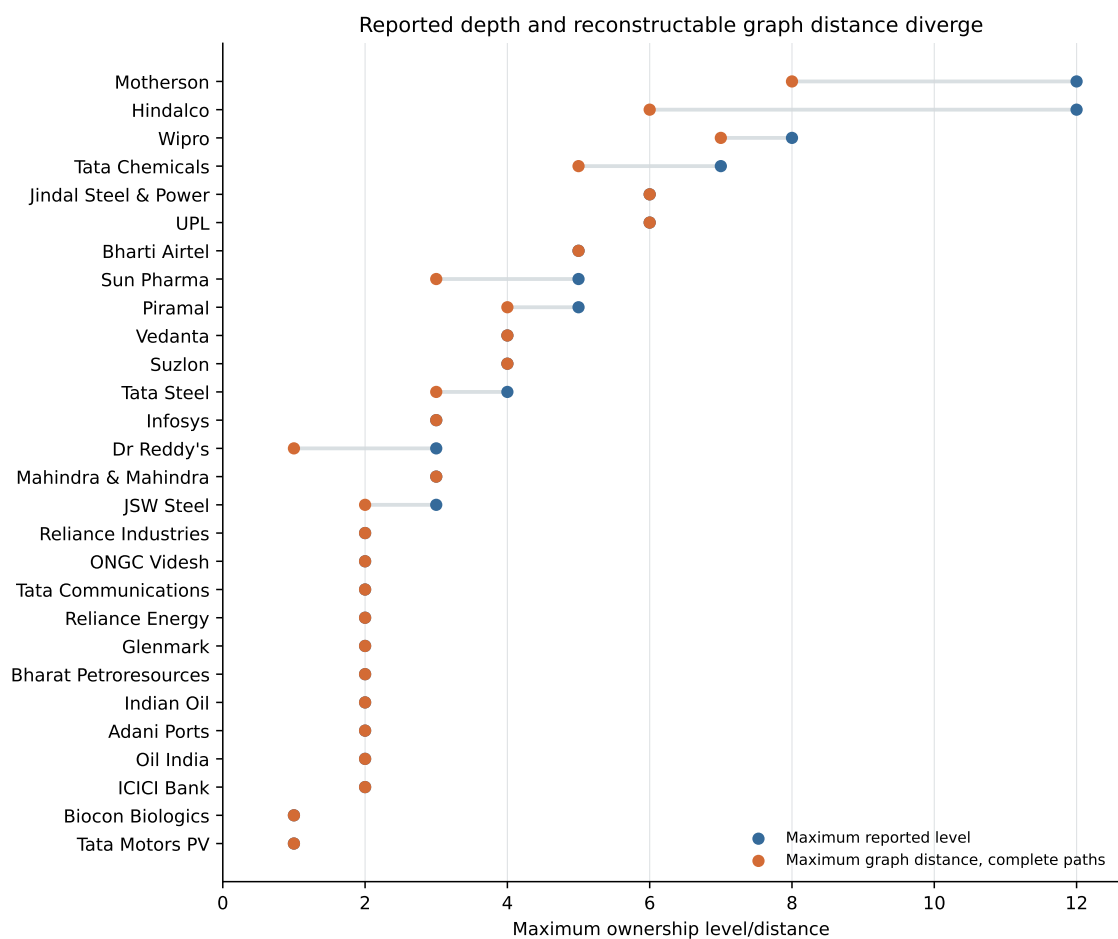


Figure 6: Maximum reported level against maximum reconstructed distance, by parent group
Notes: Each row is one parent bucket. Blue markers show maximum supplied reported level. Orange markers show maximum reconstructed distance among paths that reach the ultimate parent through observed entity nodes. The connecting segment is descriptive; neither endpoint is designated the legal truth.

8 Ownership fields, financial coverage, and structural proxies

Everything so far has used the topology—who holds whom, and where. The file also carries ownership stakes and extracted financial statements, which would in principle allow control-weighted and value-weighted versions of the same statistics. This section explains why neither is estimable here, and reports the audit as a result in its own right: it bounds what any study built on this kind of extracted evidence can claim.

8.1 Recorded stakes and the ownership-depth gradient

Stake information cannot support a complete control-weighted network. Every observed level-0 target has missing mapping stake, and 406 of 1,650 nonroot relations record exactly zero (24.6%). Where a preferred financial row contains both mapping stake and an AOC-1 shareholding measure, each comparable zero-stake case is contradicted by a positive AOC value (Appendix C). The zeros are therefore treated as unknown, not as literal absence of ownership.

Restricting to positive-stake edges produces a superficially strong gradient: the full-ownership rate is 56.4% at reported level 1 and 82.8% at level 2 or deeper, a pooled difference of 26.4 percentage points. The equal-parent difference is only 2.3 points, and excluding Reliance reduces the pooled gap to 6.1 points. The stable conclusion is a parent and portfolio composition effect, not that ownership mechanically strengthens with depth. Cumulative ownership along full paths is excluded because it would condition on positive recorded stakes and treat unresolved zeros selectively.

8.2 Financial attrition and extraction reuse

Figure 7 and Table 5 document the financial coverage funnel. A source is located for 847/1,834 targets, a balance sheet is parsed for 444, and only 265 targets are ready at least once. The preferred panel contains 560 ready target-years; 521 pass a positive-assets and nonnegative-liabilities gate, and only 105 have a testable valid profit-and-loss identity. Units are blank for 516/560 ready rows.

Duplicate-evidence checks reinforce the limitation. There are 51 repeated signatures defined jointly by fiscal year, currency, units, assets, liabilities, and equity; they cover 104 parsed rows. In 33 clusters, the same source URL is attached to different entity names and the core evidence text is byte-identical. A broad union of repeated-signature and sign flags leaves 460 ready rows, but that remainder is a diagnostic, not a validated analysis sample.

Readiness itself varies greatly across parents, and 5 parents have no ready target. The raw association between reported depth 3+ and readiness is -6.6 percentage points; a parent fixed-effect coefficient is -2.4 points, with leave-one-parent-out estimates from -6.6 to -0.8 points. This is attenuation rather than elimination, but selection, unit missingness, and extraction reuse make a financial-performance interpretation indefensible.

8.3 A limited name-based structural proxy

One supporting result survives a within-cell test. 177 targets contain “Holding(s),” “Holdco,” or “Investment(s)” in the normalized name. 64.4% of flagged targets hold at least one observed child, compared with 17.6% of other targets. Residualizing both the outcome and name flag within parent-by-country cells gives a 36.3-point association. This suggests that names can be a useful high-precision, low-recall structural proxy when edges are absent. It does not verify a legal role, and external role labels and held-out-parent validation would be required before using it as a general classifier.

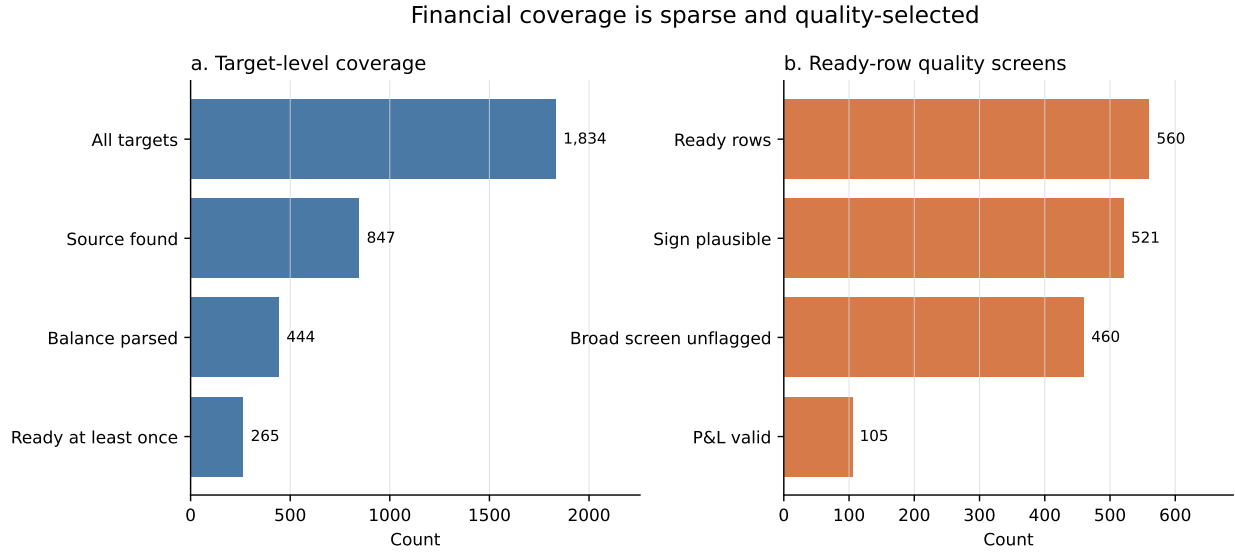


Figure 7: Target-level coverage gates and ready-row quality screens

Notes: Panel (a) counts structural targets at successive coverage gates. Panel (b) counts preferred target-year rows. The broad screen flags either a repeated numeric signature or an implausible sign combination; its unflagged remainder has not been manually validated.

Table 5: Financial coverage and quality diagnostics

Gate or diagnostic	Numerator	Denominator	Value	Unit
Targets with any source	847	1,834	46.2	targets
Targets with parsed balance sheet	444	1,834	24.2	targets
Targets ready at least once	265	1,834	14.4	targets
Ready rows	560	3,567	15.7	preferred target-years
Sign-plausible ready rows	521	560	93.0	ready rows
P&L-valid rows	105	560	18.8	ready rows
Ready rows with blank units	516	560	92.1	ready rows
Demonstrated same-URL/different-name reuse clusters	33	51	64.7	repeated-signature clusters
Ready rows unflagged by broad duplicate/sign screen	460	560	82.1	not a validated final sample

Notes: The value column is a percentage except for diagnostics whose unit states otherwise. Repeated numeric signatures are fiscal-year/currency/ units/assets/liabilities/equity clusters. Demonstrated reuse additionally requires the same URL, different target names, and byte-identical core evidence text.

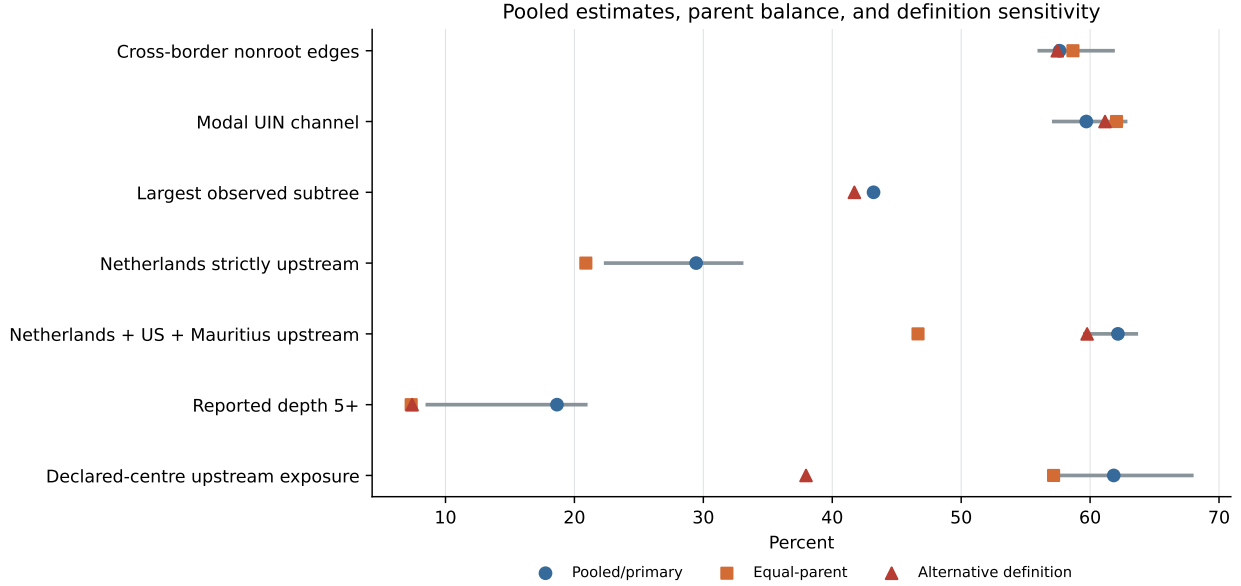


Figure 8: Each percent-scale headline estimate under four weightings and definitions
Notes: Grey segments are leave-one-parent-out ranges where applicable. Blue circles show pooled or primary estimates, orange squares show equal-parent estimates, and red triangles show the labeled alternative definition. Rows share a percent scale but not necessarily the same observation unit; Table 6 states each unit.

9 Robustness, heterogeneity, and evidence classification

The individual sections reported sensitivities as they arose. Collecting them in one place answers a different question: which of this paper’s numbers are statements about the network, and which are statements about the composition of the sample? Figure 8 and Table 6 put the main estimands on one scale for that comparison.

Three patterns are notably stable. The cross-border edge share barely moves under equal-parent weighting, raw country labels, or leave-one-parent-out checks. Modal-UIN concentration also remains high under parent balance. Dutch gateway amplification remains large under equal-parent weighting, omission of any one parent, and the no-majority-branch restriction.

Other estimates change meaningfully. The three-country upstream union and reported deep share fall under equal-parent weighting. The full-ownership depth gap shrinks sharply under parent correction. Financial readiness retains a small negative within-parent depth association, but its LOO range and severe measurement attrition preclude substantive inference. The maintained-centre exposure is much smaller when the unit is unique observed intermediary nodes rather than descendant paths.

No formal probability sample underlies the 28 parent buckets, so the paper does not present sampling-error p -values as though they supported population inference. Robustness here means sensitivity to measurement and composition: unique entities versus repeated targets, pooled versus parent- balanced weights, omission of any one parent, alternative graph definitions, fund exclusions, and explicit missingness rules.

Table 6: Canonical robustness matrix

Estimand	Unit	Primary	Parent-equal/FE	LOO range	Alternative	Definition
Cross-border nonroot edges	%	57.6	58.7	56.1 to 61.8	57.5	Raw-label sensitivity
Modal UIN channel	% of targets	59.7	62.1	57.2 to 62.8	61.2	Median parent
Largest observed subtree	median parent %	43.2	–	–	41.7	Strict DAG dominator
Dutch gateway amplification	mean descendants	34.8	37.1	21.2 to 39.7	37.8	No majority child branch
Netherlands strictly upstream	% of targets	29.4	20.9	22.4 to 33.0	–	Descendant/path estimand
Netherlands + US + Mauritius upstream	% of targets	62.2	46.7	59.6 to 63.6	59.8	Exclude venture-fund children
Reported depth 5+	% of targets	18.6	7.3	8.6 to 20.9	7.4	Reconstructed depth 5+, complete paths
Full-ownership depth gap	percentage points	26.4	2.3	–	6.1	Exclude Reliance
Financial-ready deep gap	percentage points	-6.6	-2.4	-6.6 to -0.8	–	Parent FE in equal-parent column

Notes: “Primary” is pooled unless the row labels a median-parent estimand. FE denotes a fixed-effect coefficient. LOO repeats the relevant pooled or fixed-effect estimator after dropping each parent. Alternatives are not always percentages of the same unit and should be read with the definition column.

10 Limitations and interpretation boundary

The study’s limitations are part of its empirical result rather than a generic disclaimer.

- **Sample scope.** The parent buckets are supplied, not sampled from a documented universe. Results describe these groups and cannot establish prevalence across Indian multinational firms.
- **Entity identity.** Normalized names and countries are candidate identities. They cannot distinguish every alias, reorganization, joint exposure, or legally separate same-name company without registration identifiers.
- **Hierarchy completeness.** Named missing parents truncate paths, and reported levels disagree with observed graph distance. External ownership charts are required to adjudicate which representation is legally or historically correct.
- **UIN semantics.** The data establish mapping-channel concentration but do not validate the meaning of UIN substrings or their use as dates, office codes, or ownership types. Vintage claims are therefore excluded.
- **Jurisdiction labels.** A country label is not proof of operating location, management, tax residence, or entity function. The data cannot identify a tax, treaty, financing, acquisition, governance, or operating motive.
- **Ownership measurement.** Missing level-0 stakes and contradicted zero values prevent control weighting and cumulative-ownership inference.
- **Financial selection.** Sparse coverage, blank units, mixed currencies, invalid signs, limited P&L identities, and demonstrated evidence reuse rule out population financial comparisons.
- **Time and current status.** The source combines reported structures and source years but does not validate a common current ownership date for every edge. The network is a reconstruction of the supplied report, not a claim about present legal control.

Accordingly, unusual topology is not evidence of wrongdoing. The paper does not label any company, jurisdiction, route, or transaction fraudulent, illegal, evasive, abusive, or improper. Legal, regulatory, institutional, and tax interpretations require external evidence and are outside the identified design.

10.1 What this dataset establishes, and what it cannot

Because the distinction is easy to lose in a long descriptive paper, Table 7 states it directly. The upper panel lists claims the reconstruction supports on its own terms: counts, topology, exposure under named denominators, measurement disagreements, and data-quality rates. The lower panel lists claims that look adjacent but are not identified here. Each would require an independently sourced input—company registers, dated ownership charts, a specification for the mapping identifiers, filing histories, or a documented sampling frame—and no statistic in this file substitutes for one. In particular, the frequency of a jurisdiction on ownership paths carries no information about why entities are located there.

Table 7: The evidence boundary of this study

Scope	Statement
Established by this dataset	Unit counts: source rows, preferred target-years, target occurrences, normalized entity candidates, edges, paths, UINs, and parent buckets. Reported parent-child topology and supplied country labels, subject to the documented unresolved-node and name-resolution limits. Reported level and reconstructed graph distance, reported separately, and the targets at which they disagree. Descendant counts below observed first-hop entities, by first-hop jurisdiction, under gateway, parent-equal, and omission weighting. Concentration of a parent network behind one mapping channel, one observed subtree, or one strict graph dominator. Residence, unique-intermediary, and strictly-upstream path exposure of each jurisdiction, each with its own denominator. Share of nonroot edges whose child and parent country labels differ. Internal contradictions between recorded mapping stakes and AOC-1 shareholding, and the coverage of each ownership field. Financial source, parse, readiness, sign, and P&L gates, and the reuse of identical extracted evidence across differently named targets. Association between holding-type naming and observed holder status, within parent-by-country cells.
Requires external evidence	Whether a named entity is legally identical to a same-name entity, an alias, or a jointly held vehicle. Whether reported level or reconstructed graph distance is the legally correct ownership distance. Whether UIN characters encode a registering office, an investment type, or a registration year. Whether any entity is a tax, treaty, financing, acquisition, governance, or operating vehicle. Whether a route was designed or inherited through acquisition, and whether it reflects current legal control. Whether removing a jurisdiction would detach, reroute, or tax the entities measured below it. Whether any jurisdiction belongs in a 'financial centre' category; the list used here is maintained, not observed. Whether recorded stakes measure economic ownership or control at any specific date. Whether extracted financial values are correct, comparable across currencies and units, or representative of the underlying firms. Whether the 28 supplied parent buckets represent Indian outward investment, or any other population.

Notes: The table is generated by the analysis pipeline, so the boundary travels with the results rather than with the prose. Items in the lower panel are not weaker versions of the upper panel; they are outside the design and are neither asserted nor denied here.

10.2 Questions this file cannot settle

Four questions remained open after independent reconstruction and cross-review, and each is load-bearing for at least one result.

1. **Which depth measure is correct.** Reported level and reconstructed distance disagree for 14.0% of complete paths. A reported level may count an intermediary that the file never scrapes, in which case graph distance is a lower bound; alternatively, name matching across channels may have short-circuited a chain, in which case the reported level is stale. External ownership charts are the only resolution. Every depth statement in this paper is therefore labeled by measure.
2. **What the mapping identifiers encode.** Their fixed format is visible in the data, but no supplied documentation defines the fields, so no vintage, office, or investment-type reading is used anywhere in the paper.
3. **What a recorded zero stake means.** 406 nonroot relations record exactly zero, and every comparable case with an independent shareholding field contradicts a literal zero. Until the mapping source distinguishes unknown from true zero, no control-weighted network statistic is safe.
4. **Whether an intermediary category can be measured rather than assumed.** Classifying a jurisdiction as a “centre” by a maintained list is an imposed partition; classifying it by the behavior of entities located there makes the outcome circular. The name-based proxy in Section 8 is a partial substitute, not a solution.

11 Implications for measurement

Three implications follow for anyone measuring corporate structure from data of this kind, and none of them requires an institutional assumption.

First, the registered entity is a weak proxy for the structure it heads. In this file, half the mapping channels resolve to a single foreign target while the ten largest resolve to 49.2% of all targets, and 47.5% of targets sit at least two levels below the parent. A statistic anchored on the registered entity is not simply a scaled-down version of the network statistic; the ratio between them varies by a factor of many across channels, so rescaling by an average multiplier would be misleading.

Second, entity counts and path counts answer different questions and should never be substituted for one another. The three-jurisdiction upstream union is 62.2% of target paths but only 46.7% when parents are balanced, and the maintained-centre category covers 61.8% of paths but only 38.0% of unique intermediary nodes. Both members of each pair are correct; reporting one and describing the other is not. The same discipline applies to any complexity index built from entity counts, which would here be dominated by whichever parent group happens to have the longest reported chains.

Third, a descriptive result on 28 parent groups is a statement about those groups. Most pooled statistics in this study move when parents are equally weighted, two of them by more than half their value, and single groups account for 88.1% of the depth mismatches and for most of the pooled ownership-depth gap. Leave-one-parent-out reporting is not a robustness ornament in this setting; it is the difference between a network fact and a fact about the largest firm in the sample.

12 Conclusion

The canonical reconstruction changes the object of study from a financial-source panel into an explicit set of target occurrences, normalized entity candidates, edges, and ownership paths. That correction is not cosmetic: it reverses parent rankings and prevents long source histories from masquerading as large corporate networks.

Within the supplied sample, the reported structure beneath the first foreign hop is both substantial and narrow. Mapping channels expand into networks whose median parent routes a large share of its entities through one observed company, and the concentration is similar whether it is measured by mapping channel, observed subtree, named-node-inclusive subtree, or strict graph dominator. Where the first hop lands is informative about how much follows: the Netherlands and Mauritius occupy intermediary and gateway positions that their resident counts alone would not predict, while the United States is prominent mainly as a destination. A majority of relations below the first hop cross a jurisdiction boundary, and that result barely moves under any reweighting the data permit. These are defensible facts about the mapped topology, not about purpose, legality, or current control.

The negative findings are equally important. Deep manufacturing is almost entirely a two-parent pattern. The pooled ownership-depth gradient is composition. Reported depth is not observed graph distance, and the disagreement is concentrated in one group. The financial layer cannot support comparative performance claims. A credible study of corporate ownership networks depends as much on refusing the wrong denominator, and on saying which attractive result did not survive, as on finding a large statistic.

Future work can extend the evidence boundary only with new validated inputs: official entity identifiers, dated ownership charts, documented mapping-channel semantics, verified stake fields, harmonized financial units, filing histories that distinguish inherited from constructed structure, and a transparent sampling frame. Until then, the contribution is a reproducible, falsifiable map of the reported network structure in this dataset, and an explicit account of where that map ends.

A Exact estimands and counting rules

A.1 Units

The pipeline distinguishes the following units throughout:

- a source candidate row for source and extraction audits;
- a preferred target-year for financial readiness and accounting checks;
- a target occurrence for reported path exposure;
- a parent-scoped normalized entity for within-group architecture;
- a global normalized entity for cross-parent unique-company sensitivity;
- a nonroot occurrence edge for direct child–parent geography;
- a logical entity edge for graph topology;
- a complete observed target path for reconstructed distance;
- an observed level-0 gateway for descendant amplification; and
- an ultimate-parent bucket for equal-parent and omission checks.

No table silently substitutes raw rows for entities or target paths for unique intermediaries. Percentages in generated tables always travel with a stated numerator, denominator, or unit.

A.2 Missing values and duplicates

Blank country or name fields are never imputed through fuzzy similarity. Named missing parents are kept as unresolved name–country nodes. Recorded zero stakes are treated as unknown for control analyses. Structural fields are required to be invariant within target identifier. Financial duplicates are screened using exact signatures over fiscal year, currency, units, assets, liabilities, and equity; demonstrated evidence reuse additionally requires a shared source URL, different entity names, and identical core evidence text.

A.3 Subtree and path denominators

For parent p with normalized entity set V_p and candidate observed node v , the observed-subtree share is

$$S_{pv} = \frac{|\text{Desc}(v) \cap V_p|}{|V_p| - 1}.$$

The reported parent statistic is the maximum S_{pv} over observed parent nodes. The named-node-inclusive variant expands the candidate set to unresolved named parents. The strict dominator variant removes v and counts only entity nodes that cease to be reachable from the ultimate root; alternate graph paths therefore reduce its value.

For an upstream-jurisdiction result, each terminal target contributes at most one indicator per jurisdiction, regardless of how often that jurisdiction appears on its path. A shared ancestor still appears once for every terminal target below it, which is why target-path exposure is reported separately from unique intermediary-node prevalence.

B Depth audit

Table 8: Reported depth, reconstructed distance, and parent concentration

Depth diagnostic	Numerator	Denominator	Value (%)	Unit
Reported level 2+	871	1,834	47.5	target occurrences
Reported level 5+	342	1,834	18.6	target occurrences
Reported level 5+, equal-parent	—	28	7.3	parent mean
Reconstructed distance 5+	125	1,687	7.4	complete paths
Reported/reconstructed mismatch	236	1,687	14.0	complete paths
Motherson share of mismatches	208	236	88.1	mismatching complete paths
Manufacturing targets at reported level 5+	242	508	47.6	manufacturing target occurrences
Motherson + Hindalco share of deep manufacturing	240	242	99.2	reported level-5+ manufacturing targets

Notes: Reported measures use all structural targets. Reconstructed distance and the mismatch rate use only complete observed paths. The equal-parent deep rate averages the parent-specific reported level-5+ shares.

C Ownership-field audit

Table 9: Stake coverage and ownership-depth diagnostics

Ownership diagnostic	Numerator	Denominator	Value	Unit
Level-0 stakes missing	184	184	100	level-0 targets
Nonroot recorded zero stakes	406	1,650	24.6	nonroot edges
Positive nonroot stakes	1,244	1,650	75.4	nonroot edges
Rows with mapping stake and AOC share	45	3,567	1.3	preferred target-years
Zero mapping stake but positive AOC share	7	7	100	comparable zero-stake rows
Stake/AOC difference above 1 pp	8	45	17.8	comparable rows
Full ownership at level 1	292	518	56.4	positive-stake edges
Full ownership at level 2+	601	726	82.8	positive-stake edges
Pooled level-2+ minus level-1 gap	—	1,244	26.4	percentage points
Equal-parent level-2+ minus level-1 gap	—	27	2.3	percentage points
Gap excluding Reliance	—	1,081	6.1	percentage points

Notes: Full ownership is defined as a recorded positive stake of at least 99.5 percent. Level-0 stakes are missing and nonroot zeros are not imputed. The equal-parent gap averages parent-specific rates; groups without both depth cells do not identify a paired contrast.

D Evidence disposition

Table 10: Canonical classification of discovered and reviewed findings

Class	Result	Disposition
Core	Denominator correction	Raw rows are source observations, not firms.
Core	Multidimensional parent architecture	Size, geography, depth, branching, and channels are distinct axes.
Core	Dominant channels and large observed subtrees	Report UIN, observed-subtree, named-node, and DAG estimands separately.
Core	Gateway-jurisdiction topology	Dutch and Mauritian roles survive unique-node, gateway, parent-weighted, and LOO checks.
Core	Depth measurement and parent concentration	Reported depth is not observed graph distance; pooled deep patterns are parent-driven.
Supporting	Cross-border nonroot edges	Stable under parent balance and leave-one-parent-out checks.
Supporting	Stake-zero and ownership-composition audit	Recorded zeros are unsafe; the pooled ownership-depth gradient is composition.
Supporting	Financial selection and extraction audit	Coverage and parsing quality preclude population financial conclusions.
Supporting	Holding-type name proxy	Large within parent-country association, pending external role validation.
Descriptive	Three-country descendant cover	Large path-weighted exposure but materially lower with equal-parent weighting.
Fragile	Fixed 19-jurisdiction centre exposure	Maintained list and descendant weighting determine the magnitude.
Fragile	Cumulative no-dilution claim	Positive-stake complete chains are selected and zeros are unresolved.
Unresolved	UIN substring semantics and vintage	Requires an authoritative RBI specification.
Rejected	General deep-manufacturing pattern	Two parents account for nearly all deep manufacturing targets.
Rejected	Financial health by depth or jurisdiction	Selection, units, duplicate extraction, and small valid samples preclude inference.
Unresolved	Tax, legal, regulatory, or misconduct mechanism	Topology alone does not identify purpose, legality, or causation.

Notes: “Core” requires agreement across the independent research streams plus robustness to the relevant denominator, parent-weighting, and hierarchy checks. Supporting results are defensible but not the paper’s main claim. Fragile, unresolved, and rejected claims are excluded from the core narrative.

E Reproducibility

From the repository root, the complete analysis and verification run with:

```
python -m src.analysis.run_pipeline --clean
python -m src.analysis.validate
for pass in 1 2 3; do
  pdflatex -interaction=nonstopmode -halt-on-error \
    -output-directory=paper paper/main.tex
done
```

The analysis command checks immutable input hashes and exact central invariants before writing any reported output. The validator rebuilds into a temporary directory and byte-compares every generated table, figure (PNG and PDF), the metrics file, and the generated research reports; figure PDFs are written without a creation timestamp so that they are byte-reproducible. Three LaTeX passes are required: the third resolves the table and figure numbers referenced in the text, and the compilation is warning-free. Full definitions and the output map appear in `REPLICATION.md`.